**Faculty OF COMPUTING**

**FOC DBS Meeting Minutes Held on February 02, 2024**

**In Attendance**

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| 1. Dr. Faisal Shafait (In Chair) | 13. Dr. Hashir Moheed Kiyani |
| 2. Dr. Imran Malik | 14. Dr. Muhammad Daud Abdullah Asif |
| 3. Dr. Shams Ud Din Qazi | 15. Dr. Muhammad Bilal |
| 4. Mr. Maajid Maqbool | 16. Dr. Aimal Tariq Rextin |
| 5. Dr. Shah Khalid | 17. Dr. Muhammad Moazam Fraz |
| 6. Dr. Syed Imran Ali | 18. Dr. Seemab Latif |
| 7. Dr. Mehdi Hussain | 19. Dr. Momina Moetesum |
| 8. Dr. Sana Qadir | 20. Ms.Naema Asif |
| 9. Dr. Sohail Iqbal | 21. Ms. Masabah binte islam |
| 10. Dr. Tahira Lashari | 22. Mr. Nadeem Nawaz |
| 11. Dr. Arham Muslim | 23. Ms. Iram Tariq Bhatti |
| 12. Mr. Syed Muhammad Ali Musa | 24. Ms. Mehwish Kiran |

**Meeting time: 1500 hrs. – 1600 hrs.**

| Sr# | Agenda Points / Discussion | Action By |
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| 1. | Dr Faisal Shafait welcomed the newly joined faculty member Ms. Mehwish Kiran. | |
| 2. | Dr. Shah Khalid briefed the house about Industrial FYP Intimation on PMS. As per NCEAC guidelines, we are required to have a report on FYPs collaborating with industries. Faculty members are required to use a checkbox during project creation to indicate any industry collaboration. | FoC FMs |
| 3. | Dr. Faisal commenced the meeting by outlining its purpose and addressed the following key points:<br>**a. Uniformity in Course Delivery:**<br>Dr. Faisal emphasized the need for consistent vigor in course delivery across all sections of a course. It was highlighted that the responsibility lies with the KGH to ensure this consistency. To achieve this:<br>● KGH is tasked with convening a meeting with all Faculty Members (FMs) at the onset of each semester to ensure uniformity in course outlines. The meeting should take place within the first week of the semester commencement.<br>● KGH must ensure that the depth of course content remains consistent across all sections of the same course.<br>● KGH must ensure the course outline of UG and PG courses are according to the degree level, PG course content should not be a repetition of the corresponding UG course.<br>**b. Standardization of Examinations:**<br>Dr. Faisal stressed the importance of maintaining a balanced level of difficulty in examinations. It was proposed that:<br>● Difficulty level and length of the exam papers should be consistent for all courses to ensure fairness. Exam paper vetting by the KGH has to take place one week prior to the commencement of the exam week. | Knowledge Group Head (KGH) |

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| | ● FMs teaching different sections of the same course across various degree programs should collaborate in setting exam papers to maintain consistency.<br>● Each Faculty Member (FM) will be required to submit an undertaking on a Google Form. Within this undertaking, the FM will confirm that the following:<br>i. The question paper they have prepared is distinct from those used in the two preceding offerings of the course. It was determined that the similarity between question papers should not exceed 30%.<br>ii. The exam has been vetted by the KGH.<br>**c. Coherence of Teaching Content:**<br>Dr. Faisal emphasized the significance of coherence in teaching content, particularly for courses that are offered in a particular sequence (e.g. Artificial Intelligence -> Deep Learning; Data Structures & Algorithms -> Design and Analysis of Algorithms. It was decided that:<br>● KGH should ensure that teaching content is coherent and avoids repetition across sequence courses. | |
| 4. | Dr. Arham highlighted the importance of coherence and standardization in assessment across all courses. Dr. Faisal further elaborated on the topic, emphasizing that ensuring standardization in assessment falls under the purview of the respective Heads of Departments (HoDs).<br>**a. Midterm Review Results Submission:**<br>Dr. Faisal approved the submission of midterm review results by each Faculty Member (FM) in the 10th week of the semester to their respective HoDs. Hence, a Mid-Semester DBS for interim result review will take place.<br>**b. Responsibility of HoDs:**<br>It was agreed that HoDs are responsible for ensuring that both lab and course component assessments are conducted meticulously and meet the required criteria. For each lecture, CH, at least one assignment and two quizzes are expected.<br>**c. Responsibility of FMs and LEs:**<br>Each FMs and LEs are responsible for submitting their assessment to their course HoD in the 10th week of each semester. | KGH and respective course program HoDs<br><br>FMs and LEs |
| 5. | House showed serious concerns about the lab's component assessment.<br>**a. Lab Engineer Responsibilities:**<br>All labs should be completed within the stipulated time of the lab. LEs should actively engage with the students and not allow any student to leave the lab unless he/she has submitted the lab tasks. They should monitor the student's activities by keeping an eye on their screens to discourage the use of unfair means (e.g. ChatGPT / copying code from peers).<br>b. **Responsibility of Faculty Members (FMs):**<br>The House decided that lab component assessment is the responsibility of the respective FMs. Therefore, it was determined that each lab engineer should get lab evaluation results reviewed by the respective FM.<br>**c. Implementation of Quizzes:** | Lab engineers and Faculty Members |

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| | To enhance assessment effectiveness, it was decided that in the last 15 minutes of each lab session, a 5-10 minute quiz on the same day lab tasks will be conducted on the Learning Management System (LMS) by the lab engineer. It was further stipulated that quiz questions must be designed by the lab engineer and duly approved by the respective FM prior to conduct. | |
| 6. | **Student Advisory:**<br>● Student Advisors are advised to make sure to keep records of all the correspondence with the students.<br>● Report the Students who are not attending advisory sessions to ACB. | FoC FMs |
| 7. | **Class Audits:**<br>● All class auditors are instructed to ensure the quality of class audits. | FoC FMs Class auditors |
| 8. | **Course Outline Approval:**<br>• LQEC presented the updated course outline for CS212 OOP.<br>• The House approved the changes. Updated outline is attached in Annex A. | |
| 9. | The department coordinator briefed the house about the new TA hiring policy | |



FOC Department Coordinator

February 12, 2024

**Copy to:**

Office DOC

Principal SEECS

Annex-A

# COURSE OUTLINE

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| **Department:** | Faculty of Computing | **Knowledge Group:** | KG-PC | |
| **Programme:** | Computer Science (BSCS) | **Class:** | BESE-2K23, BSCS-2K23 | |
| **Course code:** | CS-212 | **Academic Session/Semester:** | | Spring 2024 |
| **Course name:** | Object Oriented Programming | **Pre/co requisite (course name and code, if applicable):** | | CS-110 FoCP |
| **Credit hours:** | 3+1 | | | |

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| **Course Synopsis** | This course will introduce the Object-Oriented (OO) philosophy to software development, which is a modern and powerful approach. Today, many programming languages support the OO concepts. However, during this course, we'll majorly use the Java programming language.<br>The objective is to make the students understand the benefits of using OO techniques over procedural programming practices, and thereby motivating them to use OO concepts in software development. Further, the students will learn some powerful features of the Java programming language. | | | |
| **Course Learning Outcomes (CLOs)** | 1. Understand core OOP concepts such as classes, objects, encapsulation, inheritance, and polymorphism.<br>2. Evaluate complex programming problems using object-oriented principles analysing solutions for efficiency and scalability.<br>3. Build programs using the latest IDEs as per standard practices applicable to the software industry.<br>4. Develop solutions to real-world problems by applying appropriate object-oriented techniques.<br>5. Demonstrate commitment to professional ethics by following engineering norms applicable to the software industry. | | | |
| **Course Schedule** | Monday 10:00-10:50<br>Tuesday 10:00-10:50<br>Thursday 15:00-15:50<br>Lab (Tuesday 14:00-16:50) | | | |
| **Course lecturer** | **Name** | **Office** | **Contact no.** | **E-mail** |
| | Mehwish Kiran | A-208 | | mehwish.kiran@seecs.edu.pk |

**Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:**

| No. | Course Learning Outcomes | PLO (SE) | PLO (CS) | BT Level | Teaching & Learning Methods | Assessment Methods |
|---|---|---|---|---|---|---|
| CLO 1 | **Understand** core OOP concepts such as classes, objects, encapsulation, inheritance, and polymorphism. | PLO-1 | PLO-1 | C-2 (Understand) | Active learning, Cooperative Learning, Blended Learning | Assignment<br>Quiz<br>MSE<br>ESE |
| CLO 2 | **Evaluate** complex programming problems using object-oriented principles analysing solutions for efficiency and scalability. | PLO-2 | PLO-3 | C-5 (Evaluate) | Active learning, Cooperative Learning, Blended Learning | Assignment<br>Quiz<br>MSE<br>ESE<br>Labs |
| CLO 3 | **Develop** solutions to real-world problems by applying appropriate object-oriented techniques. | PLO-3 | PLO-4 | C-6 (Create) | Active learning, Cooperative Learning, Blended Learning | MSE<br>ESE<br>Assignments<br>Project<br>Labs |

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| CLO 4 | **Build** programs using the latest IDEs as per standard practices applicable to the software industry. | PLO-5 | PLO-5 | P-3 (Guided Response) | Active learning, Cooperative Learning, Blended Learning | Project Labs |
| CLO 5 | **Demonstrate** commitment to professional ethics by following engineering norms applicable to the software industry. | PLO-8 | PLO-9 | A-3 (Valuing) | Active learning, Cooperative Learning, Blended Learning | Project Labs |

**Details on Innovative T&L practices:**

| No. | Type | Implementation |
|---|---|---|
| 1. | Active learning | Conducted through in-class or lab activities. |
| 2. | Cooperative learning | Conducted through design project. Students in a team of five will be given a design project that requires software engineering process design solutions. Compliance to the design specifications need to be given in the form of written reports. |
| 3. | Blended learning | Conducted through Learning Management System (LMS) of NUST. All information as well as materials related to teaching and learning activities will be shared with class through this system. Some for formative assessments will be also conducted using this system. |

**Weekly Schedule:**

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| Week 1 | Java Basics<br>• Introduction to Java Programming Language<br>• Compiling & Executing Simple Java Programs<br>• Data types & Operators<br>• Input / Output |
| Week 2 | Flow Control & Composite Datatypes<br>• Decisions<br>• Loops and Iterations |
| Week 3 | Arrays and strings / functions |
| Week 4<br>Week 5<br>Week 6 | Classes and Objects<br>• Basic Concepts<br>• Encapsulation & Data Hiding<br>• Access modifiers<br>• Constructors & Destructors<br>• Setters & Getters<br>• Copy Constructor<br>• Default and No-Argument Constructors<br>• Method (Function) Overloading<br>• Static Class Members<br>• this Reference |
| Week 7<br>Week 8 | Inheritance In Java<br>• Super classes and sub classes<br>• Protected members<br>• Method (Function) Overriding<br>• Constructors in subclasses |
| **Week 9** | **Mid-Semester Break** |
| Week 10<br>Week 11 | Polymorphism<br>• Abstract Classes & Methods<br>• Polymorphic Behaviour |

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| | • Final Methods and Classes |
| Week 12 | Interfaces and Abstract Classes |
| Week 13 | Exception Handling |
| Week 14 | Graphical User Interfaces |
| Week 15 | File Manipulation |
| Week 16 | Project Presentations |
| Week 17 | Revisions |
| **Week 18** | **End Semester Break** |

**Lab Experiments (if applicable):**

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| Lab 1 | Java Basics [Data types and Operators] |
| Lab 2 | Flow Control [Loops and Decisions] |
| Lab 3 | Composite data types [Arrays & Strings] |
| Lab 4 | Classes and Objects |
| Lab 5 | Constructors |
| Lab 6 | Composition |
| Lab 7 | Inheritance |
| Lab 8 | Method Overriding |
| **Week 9** | **Mid-Semester Break** |
| Lab 9 | Polymorphism |
| Lab 10 | Interfaces and Abstract Classes |
| Lab 11 | Exception Handling |
| Lab 12 | GUI |
| Lab 13 | Event Handling |
| Lab 14 | I/O and File Handling |
| Lab 15 | Open-Ended Lab |
| Lab 16 | Advanced Topics |
| **Week 18** | **End Semester Break** |

**Assessment Methods:**

| Assessment | | Percentage |
|---|---|---|
| **Theory: 75%** | | |
| 1 | Quizzes (10-15%) | 15% |
| 2 | Assignments (5-10%) | 10% |
| 3 | Mid-Term Exam (25-35%) | 30% |
| 4 | End-Semester Exam (40-50%) | 45% |
| **Labs:25%** | | |
| 1 | Labs | 70% |
| 2 | Project / Final Lab | 30% |
| **Total:** | | 100 |

**Learning resources:**

**Textbook:**

1. Java – How to Program (Early Objects), Paul Deitel and Harvey Deitel, 10th Edition, ISBN 978-1- 292-01819-5, Pearson Education, 2015
2. Bruce Eckel, Thinking in Java, Fourth Edition, ISBN-13: 978-0131872486, 25th March, 2006. (available online)

**Reference Book:**

1. Computing Concepts with Java Essentials, Cay Horstmann, 3rd Edition, ISBN 0-471-24371-x, 2003, John Wiley & Sons.
2. Object Oriented Programming in C++, Robert Lafore, 4th Edition, ISBN-10: 0672323087 | ISBN-13: 978-0672323089, 2001

**Grading Policy:**

**Quiz Policy:**

The quizzes will be unannounced / announced and normally last for ten minutes. The question framed is to test the concepts involved in last few lectures. Number of quizzes that will be used for evaluation is at the instructor's discretion.

**Project Policy:**

Students will be required to develop a project during the course which should be completed towards the end of the semester. They will be graded based on project deliverables and presentation at the end. Students will work in a group/team for projects. A group of 3 students is recommended. At most 4 students are allowed.

**Assignment Policy:**

In order to develop comprehensive understanding of the subject, assignments will be given. Late assignments will not be accepted / graded. All assignments will count towards the total (No 'best-of' policy). The students are advised to do the assignment themselves. Copying of assignments is highly discouraged and violations will be dealt with severely by referring any occurrences to the disciplinary committee. The questions in the assignment are meant to be challenging to give students confidence and extensive knowledge about the subject matter and enable them to prepare for the exams.

**Class participation:**

The students are encouraged to participate in class by actively taking part in asking questions from the instructor, sharing his/her thoughts about the topic under discussion, replying to instructor questions, contribute in project presentation and demo. The class participation will be recorded by the instructor and 2% of project marks are assigned to student class participation.

**Plagiarism:**

SEECS maintains a zero-tolerance policy towards plagiarism. While collaboration in this course is highly encouraged, you must ensure that you do not claim other people's work/ ideas as your own. Plagiarism occurs when the words, ideas, assertions, theories, figures, images, programming codes of others are presented as your own work. You must cite and acknowledge all sources of information in your assignments. Failing to comply with the SEECS plagiarism policy will lead to strict penalties including zero marks in assignments and referral to the academic coordination office for disciplinary action.